

HITARTH CHOPRA

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[Portfolio](#) | [Leetcode](#) | [Codeforces](#)

EDUCATION

Texas A&M University | Master of Computer Science | GPA: 3.9/4.0 Expected May 2026

Relevant Courses: Parallel Computing, Reinforcement Deep Learning, Algorithms, Information Storage & Retrieval

Delhi Technological University | Bachelor of Technology | GPA: 9.02/10 August 2020 - May 2024

Relevant Courses: Database Management, Object Oriented Programming, Computer Architecture, Operating Systems

EXPERIENCE

Data Scientist | Texas A&M Engineering Experiment Station December 2024 - April 2026

- Designing synthetic electric grid models using **algorithmic optimization & fuzzy machine learning** from WECC data
- Engineered **optimized SQL-based ETL pipelines** for the EIA-860, HRRR, and OSM datasets by query parallelization, indexing strategies, and caching layers to reduce data retrieval overhead **cutting monthly costs by \$8,000**

Software Developer Intern | Teesta Investment February 2024 - April 2024

- **Automated multi-exchange trades** using cross-platform trading strategies across 5 exchanges for **real-time hedging**
- Engineered **low-latency, multi-threaded trading bots** in Python deployed on **Linux-based servers** and AWS, leveraging containerized microservices for real-time trade execution under **37ms latency**.

Front-End Developer Intern | Whizlabs Software June 2023 - February 2024

- Revamped the Task Tracker UI/UX optimizing render performance for **1,000,000+ users**
- Led the team in building a **notification alert system** with a React-Redux / Node.js-Express stack through leveraging asynchronous state management and optimized REST API pipelines, reducing schedule collisions by **8%**

Student Developer | Morgan Stanley May 2023 - June 2023

- Designed a web application for **analyzing the biodata of 1600+ children** at Bal Asha Trust (child orphanage trust)
- Implemented **multi-lingual support** in 10+ languages and developed a pipeline for children's publicity in **1 day**

ACADEMIC PROJECTS

News Classifier Model | [Code](#) | Deep Learning, Adversarial NLP August 2024 - October 2024

- Developed a **LLM-based** pipeline for feature visualization & content drift across **40,000+** news & twitter articles
- Incorporated a **sentiment-analysis framework** for a novel fusion of linguistic retrieval & deep learning (**F1 = 0.89**)

Ant Colony Optimization Simulator | [Code](#) | [Website](#) | Graph Theory, Javascript May 2023 - June 2023

- Simulated the popular **population-based metaheuristic** Ant Colony System algorithm with 4 different speeds
- The simulator generated approximate **solutions for optimization problems**, such as the traveling salesman problem

Electromagnetic Field Detector | [Code](#) | [Hardware](#) | C++, Arduino IDE May 2022 - June 2022

- Constructed hardware using 10 analog pins to detect the **relative magnitude** of EMF radiations in electronic proximity
- Enhanced the model to determine the EMF magnitude by wiring in an LED screen to **find a faulty device in 3 seconds**

ACHIEVEMENTS

- Secured a rating of **2185** on **Leetcode**, **Top 0.5%** with best contest rating of **416** among **24,000** participants
- Achieved **1600+ rating (Top 1%)** on **Codeforces** with best contest rating of **334** among **14,000** participants
- Promoted to **Senior Mentor** at 'Roundtable DTU' to help a team of **100+ students** in Data Structures and Algorithms
- Appointed **Student Mentor** for 'Desh ke Mentor' to provide career counseling for **16 underprivileged students**

RESEARCH AND PUBLICATIONS

[Comprehensive Methodology for Sanitizing Geographic Data in Electric Grid Models](#) PI: Dr. Jonathan Snodgrass

Engineered an **automated geospatial ETL** and **graph-validation pipeline** for large-scale power grid models (EIA-860)

[Analysis of Solar Energy Mini-Grids for Rural Electrification](#) PI: Dr. JP Kesari

Engineered a **scalable solar mini-grid modeling pipeline** for rural electrification and energy reliability analysis

TECHNICAL SKILLS

Languages and Frameworks: Python, C++, C, Javascript, AWS, C#, Java, React, Kotlin, Tensorflow, Angular, NodeJs

Tools and Platforms: LLM, CSS, Git, PyTorch, ExpressJs, Pandas, Power BI, Tableau, NumPy, Docker, Kubernetes, JAX

Databases and Operating Systems: MongoDB, PostgreSQL, Oracle, MySQL, Unix, Ubuntu, MacOS, Linux, Bash